ECON2050 Tutorial 2

Basic Real Analysis II

Semester 2 2026

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Announcement

- Assignment 1 has been released.
- Due date: 4/09/2026 4:00 pm.
- Worth 16% of your final grade!
- I'll go through some important points for the assignment at week 5 tutorial.
- Feel free to come to my consultation.
  - My consultation hours are every Thursday from 12:30 pm to 2:00 pm, in 39-125A.

Today's Plan

- Recap: key ideas from Lecture 2
- Tutorial questions: Q1–Q5. Hopefully, we can cover all of them...

Recap: Functions of Several Variables

1D so: $f: \mathbb{R} \to \mathbb{R}$

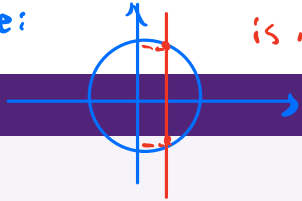
General form

A function of several variables can be written as $f: D \to \mathbb{R}^m$, where the domain is some set $D \subseteq \mathbb{R}^k$.

$\mathbb{R}^k$ $\mathbb{R}^m$

Note. A function assigns each input exactly one output!!!

Example:



is not

Graph

For $f: D \subseteq \mathbb{R}^2 \to \mathbb{R}$, the graph is

$$\text{graph}(f) = \{(\boldsymbol{x}, y) \in D \times \mathbb{R} \subseteq \mathbb{R}^3 : y = f(\boldsymbol{x})\}.$$

Level curve

The level curve of level c is

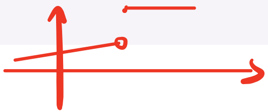
$$\{\boldsymbol{x} \in D : f(\boldsymbol{x}) = c\}.$$

A contour plot shows level curves for several values of c.

Recap: Continuity

Sequential definition

Let $D \subseteq \mathbb{R}^k$. A function $f: D \to \mathbb{R}^m$ is **continuous at $\boldsymbol{x} \in D$** if for every sequence $(\boldsymbol{x}_n)$ in D such that $\boldsymbol{x}_n \to \boldsymbol{x}$, we have $f(\boldsymbol{x}_n) \to f(\boldsymbol{x})$.



Note.

- You may have seen other definitions of continuity. They are all equivalent.
- Intuition: continuity means a function behaves smoothly, without any sudden jumps.

Continuity on a domain

f is **continuous** if it is continuous at every $\boldsymbol{x} \in D$.

Recap: Tools for Proving Continuity

Algebra of continuous functions

If $f: D \to \mathbb{R}$ and $g: E \to \mathbb{R}$ are continuous at $\boldsymbol{x} \in D \cap E$, then

$$f + g, \quad f - g, \quad f \cdot g,$$

and f/g when $g(\boldsymbol{x}) \neq 0$, are continuous at $\boldsymbol{x}$.

Composition

If $f: D \to \mathbb{R}^m$ is continuous at $\boldsymbol{x}$ and $g: E \to \mathbb{R}^\ell$ is continuous at $f(\boldsymbol{x})$, with $f(D) \subseteq E$, then

$$g \circ f: D \to \mathbb{R}^\ell, \qquad \boldsymbol{x} \mapsto g(f(\boldsymbol{x}))$$

is continuous at $\boldsymbol{x}$.

Example: $\boxed{h(x)} = \sin(x^2)$ $\quad f(x) = x^2$ is cts

$= g(f(x))$ $\quad g(x) = \sin x$ is cts

Recap: Vector-Valued Functions and Projections

Componentwise continuity

Let $f = (f_1, \ldots, f_m) : D \to \mathbb{R}^m$. Then f is continuous at $\boldsymbol{x} \in D$ if and only if every component function f_i is continuous at $\boldsymbol{x}$.

Projection functions

For each $i = 1, \ldots, k$, the projection

$$\text{pr}_i : \mathbb{R}^k \to \mathbb{R}, \qquad \boldsymbol{x} \mapsto x_i$$

is continuous.

Example: $f : \mathbb{R}^2 \to \mathbb{R}$, $f(x,y) = x^2 = h \circ g$ is cts

$g : \mathbb{R}^2 \to \mathbb{R}$, $\text{pr}(x,y) = x$ is cts

$h = x^2$ is cts

continuous

Recap: Linear and Affine Functions

Inner product

$$\text{For } \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^k, \;\; \boldsymbol{x} \cdot \boldsymbol{y} = \sum_{i=1}^{k} x_i y_i = x_1 y_1 + \cdots + x_k y_k.$$

Linear and affine functions

$\mathbb{R}^k \to \mathbb{R}$ $\quad f(x) = a \cdot x$

- A function $f : \mathbb{R}^k \to \mathbb{R}$ is **linear** if there exists $\boldsymbol{a} \in \mathbb{R}^k$ such that $f(\boldsymbol{x}) = \boldsymbol{a} \cdot \boldsymbol{x}$.
- It is **affine** if there exist $\boldsymbol{a} \in \mathbb{R}^k$ and $b \in \mathbb{R}$ such that $f(\boldsymbol{x}) = \boldsymbol{a} \cdot \boldsymbol{x} + b$. $\quad b = 0$

$\in \mathbb{R}$

Change formula

For an affine function $f(\boldsymbol{x}) = \boldsymbol{a} \cdot \boldsymbol{x} + b$, if $\Delta \boldsymbol{x} = \boldsymbol{x} - \boldsymbol{x}_0$, and ① $\Delta y = f(\boldsymbol{x}) - f(\boldsymbol{x}_0)$, then

$\mathbb{R}^2$: $\Delta y = a \cdot \Delta x$

② $$\Delta y = \boldsymbol{a} \cdot \Delta \boldsymbol{x} = \sum_{i=1}^{k} a_i \Delta x_i.$$

$= a_1 \cdot x_1 + a_2 \cdot x_2$

Question 1

Let f denote an arbitrary function $f : \mathbb{R}^2 \to \mathbb{R}$. For each statement, determine whether it is true or false. If false, provide an explanation or a counterexample.

(a) If $f(x_0, y_0) = f(x_1, y_1)$, then $x_0 = x_1$ and $y_0 = y_1$.

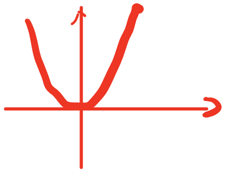
(b) If $a \in \mathbb{R}$, then $f(ax, ay) = a^2 f(x, y)$.

(c) A vertical line, parallel to the z-axis, can cross the graph of f at most once.

(d) Two level curves of f can intersect.

(a) If $f(x_0, y_0) = f(x_1, y_1)$, then $x_0 = x_1$ and $y_0 = y_1$.

one-to-one function



False. It depends on the function form.

Counterexample: $f(x,y) = xy$. $\begin{cases} (1,1) : f(1,1) = 1 \cdot 1 = 1 \\ (-1,-1) : f(-1,-1) = (-1) \cdot (-1) = 1 \end{cases}$

$1, 3$

(b) If $a \in \mathbb{R}$, then $f(ax, ay) = a^2 f(x, y)$. Homogeneous of degree 2 function

 1

 3

False. It depends on the function form.

Counterexample: $f(x,y) = x+y$

$f(ax,ay) = ax + ay = a(x+y) = a^1 f(x,y)$: Homogenous of degree 1 function

(c) A vertical line, parallel to the z-axis, can cross the graph of f at most once.

$f: \mathbb{R}^2 \to \mathbb{R}$. A function assigns each input exactly one output.

True.

(d) Two level curves of f can intersect. $\Rightarrow$ at (x_0,y_0), it has two level curves $\Rightarrow$ it has two output

Level curve of level k: $\{(x,y) \in \mathbb{R}^2 : f(x,y) = k\}$

False. Proof by contradiction.

Proof. Suppose there are two level curves $\{x : f(x) = c_1\}$ and $\{x : f(x) = c_2\}$ with $c_1 \neq c_2$ and those two level curve intersect at (x_0, y_0).

$f(x_0, y_0) = c_1$

$f(x_0, y_0) = c_2$

$\Rightarrow f(x_0, y_0) = c_1 = c_2$. Contradiction!

The statement is false

$\square$

Tutorial Q1: True or False

Knowledge used

- A function assigns each input exactly one output.
- The graph of $f : \mathbb{R}^2 \to \mathbb{R}$ contains exactly one height $z = f(x, y)$ above each input (x, y).
- A level curve of level C is $\{(x, y) : f(x, y) = C\}$.

Solution

(a) False. Let $f(x, y) = xy$, $(x_0, y_0) = (1, 2)$ and $(x_1, y_1) = (-1, -2)$. Then $f(1, 2) = 2 = f(-1, -2)$, but the two inputs are different.

(b) False. Let $f(x, y) = x + y$. Then

$$f(ax, ay) = ax + ay = a(x + y) = af(x, y),$$

which is not generally equal to $a^2 f(x, y)$.

(c) True. For each (x, y), the function gives a unique value $z = f(x, y)$.

(d) False. If two distinct level curves $C_1 \neq C_2$ intersected at (x, y), then $f(x, y) = C_1$ and $f(x, y) = C_2$, a contradiction.

Question 2

Recall that a level curve of a function of two variables f is a subset of f's domain whose elements are the points (x, y) for which, for some $k \in \mathbb{R}$, $f(x, y) = k$.
Determine the domain on which the following functions can be defined and draw a few level curves for each.

$f: \mathbb{R}^2 \to \mathbb{R}$

Restriction : Square root : $\sqrt{}$ ≥ 0
Fraction : $\frac{}{\neq 0}$
$\ln(\cdot)$: > 0

(a) $f(x, y) = (x-1)^2 + (y+2)^2$.

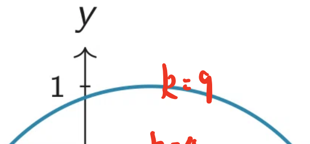
(b) $f(x, y) = x + y^2 + 1$.

(a) $f(x,y) = (x-1)^2 + (y+2)^2$. Domain $= \mathbb{R}^2$

Level curve of level k : $\{(x,y) \in \mathbb{R}^2 : f(x,y) = (x-1)^2 + (y+2)^2 = k\}$

$k=1$: $(x-1)^2 + (y+2)^2 = 1 \Rightarrow r = 1$

$k=4$: $(x-1)^2 + (y+2)^2 = 4 \Rightarrow r = 2$



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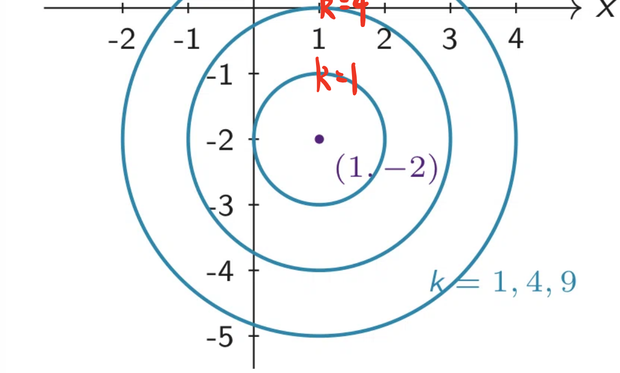
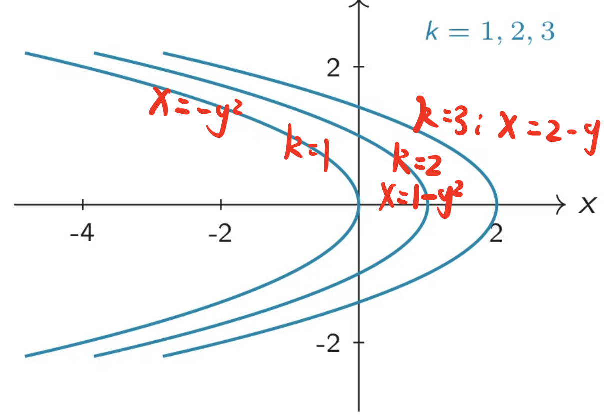




(b) $f(x, y) = x + y^2 + 1.$

Domain $= \mathbb{R}^2$. Level curve of level k : $\{(x,y) \in \mathbb{R}^2 : f(x,y) = x+y^2+1 = k\}$

Parabola: $x = k-1-y^2$

Replace x with y

$y = k-1-x^2$

$k=1$: $x+y^2+1 = 1 \iff x = -y^2$

$k=2$; $x+y^2+1 = 2 \iff x = 1-y^2$

$k=3$; $x+y^2+1 = 3 \iff x = 2-y^2$

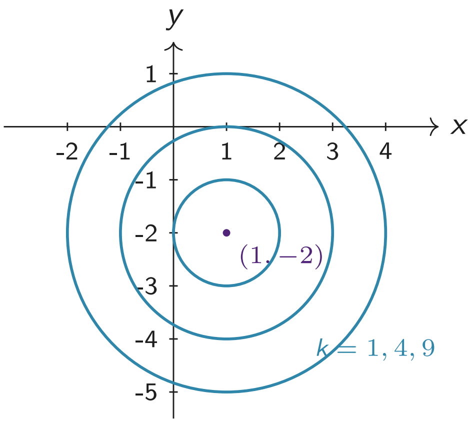
Tutorial Q2(a): Domain and Level Curves

Knowledge used

- The domain is the largest set on which the formula is well-defined.
- A level curve of level k solves $f(x, y) = k$.

Solution

For
$$f(x, y) = (x - 1)^2 + (y + 2)^2,$$
the domain is
$$D = \mathbb{R}^2.$$
The level curve of level k is
$$(x - 1)^2 + (y + 2)^2 = k.$$
For $k \geq 0$, this is a circle centered at $(1, -2)$ with radius $\sqrt{k}$.



Tutorial Q2(b): Domain and Level Curves

Knowledge used

- Level curves are found by setting $f(x, y) = k$ and solving for a geometric relation.
- Different values of k shift the level curve.

Solution

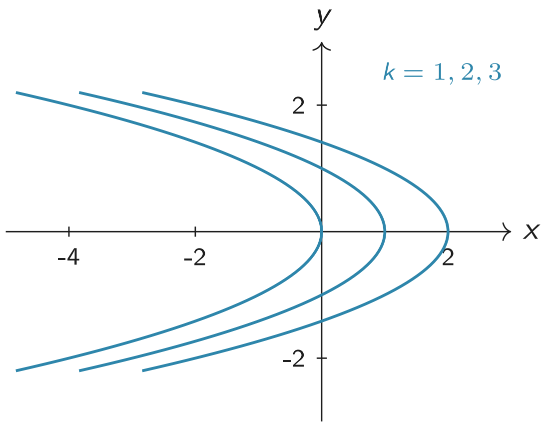
For $f(x, y) = x + y^2 + 1$, the domain is

$$D = \mathbb{R}^2.$$

The level curve of level k is

$$x + y^2 + 1 = k \iff x = k - 1 - y^2.$$

For $k = 1$, this is $x = -y^2$, which passes through the origin.



Tutorial Q3

Question 3

Recall that $\ln(\cdot)$, $\sin(\cdot)$, and $\cos(\cdot)$ are continuous. Prove that the following functions are continuous.

(a) $f : \mathbb{R} \to \mathbb{R}$, $x \mapsto (\cos x)^2$.

(b) $f : \mathbb{R}^2 \to \mathbb{R}^2$, $(x, y) \mapsto \left(\ln(x^4 + 1), \ln(y^4 + 1)\right)$.

(c) $f : \mathbb{R}^2 \to \mathbb{R}$, $(x, y) \mapsto \sin(x^2) + \sin(y^2)$.

Algebra: $f+g$, $f-g$, $f \cdot g$ is cts

Composition: $h = \underbrace{f}_{\text{cts}} \circ \underbrace{g}_{\text{cts}} \Rightarrow h$ is cts

$f = (f_1, f_2)$ is cts $\Leftrightarrow \begin{cases} f_1 \text{ is cts} \\ f_2 \text{ is cts} \end{cases}$

$pr(x, y)$ is cts

(a) $f : \mathbb{R} \to \mathbb{R}$, $x \mapsto (\cos x)^2$.

$g : \mathbb{R} \to \mathbb{R}$, $g(x) = \cos x$ is cts

$h : \mathbb{R} \to \mathbb{R}$, $h(x) = x^2$ is cts

$f = h(g(x)) \Rightarrow$ is cts

Proof: Let $g(x) = \cos x$ and $h(x) = x^2$. Then $f(x) = h(g(x))$.
Because $\cos x$ and x^2 are cts, their composition f is also cts.
$\square$

(b) $f : \mathbb{R}^2 \to \mathbb{R}^2$, $(x, y) \mapsto (\ln(x^4 + 1), \ln(y^4 + 1))$.

WTS $f = (f_1, f_2)$ is cts $\Leftrightarrow \begin{cases} f_1 \text{ is cts} \\ f_2 \text{ is cts} \end{cases}$

$f_1 : \mathbb{R}^2 \to \mathbb{R}$, $f_1(x, y) = \ln(x^4 + 1)$

$pr_1 : \mathbb{R}^2 \to \mathbb{R}$, $pr(x, y) = x$ is cts

$g : \mathbb{R} \to \mathbb{R}$, $g(x) = x^4 + 1$ is cts

$h : \mathbb{R}_{++} \to \mathbb{R}$, $h(x) = \ln x$ is cts

$f_1(x, y) = h(g(pr_1(x, y))) \Rightarrow$ is cts

Similarly, $\underbrace{f_2 = h(g(pr_2(x, y)))}_{\text{cts}}$, where $pr_2(x, y) : \mathbb{R}^2 \to \mathbb{R}$, $pr_2(x, y) = y$.

Proof. Let $f = (f_1, f_2)$, where $f_i = h(g(pr_i(x, y)))$, $i = 1, 2$. $g(x) = x^4 + 1$, $h(x) = \ln(x)$.
Because pr_i, $x^4 + 1$, and $\ln x$ are cts, the composition f_i is cts.
Since each component of f is cts, we can conclude f is cts.
$\square$

Tutorial Q3: Continuity Proofs

Knowledge used

- Use composition, sums, and componentwise continuity.
- Projection maps $\text{pr}_i(\boldsymbol{x}) = x_i$ are continuous.

Solution

(a) Let $g : \mathbb{R} \to \mathbb{R}$, $t \mapsto t^2$. Then $f = g \circ \cos$, so f is continuous.

(b) Write $f = (f_1, f_2)$, where

$$f_i(\boldsymbol{x}) = \ln(x_i^4 + 1) = \ln \circ g \circ \text{pr}_i(\boldsymbol{x}), \qquad g(t) = t^4 + 1.$$

Each component is continuous; hence f is continuous.

(c) With $g(t) = t^2$,

$$f = \sin \circ g \circ \text{pr}_1 + \sin \circ g \circ \text{pr}_2 .$$

Therefore, f is continuous.

Tutorial Q4

Question 4

The level curves of a production function $Q(K, L)$ are called isoquants. Obtain and plot a few isoquants for the following production functions.

(a) $Q(K, L) = K^{\frac{1}{2}} L^{\frac{1}{2}}$.

(b) $Q(K, L) = 0.5\ln(K) + 0.5\ln(L)$.

Positive Real Number

Domain : $\mathbb{R}^2_{++}$

$\hookrightarrow$ positive

$e^{\ln x} = x$

(b) $Q(k, L) = \frac{1}{2}\ln(k) + \frac{1}{2}\ln(L)$

level curve of level c is $\{(k, L) \in \mathbb{R}^2_{++} : \frac{1}{2}\ln(k) + \frac{1}{2}\ln(L) = c\}$

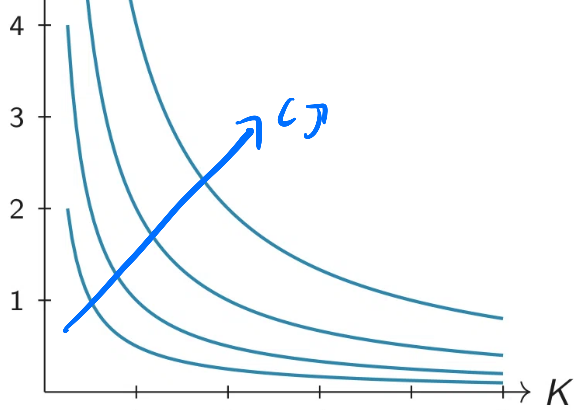
$\frac{1}{2}\ln(k) + \frac{1}{2}\ln(L) = c \implies e^{\frac{1}{2}\ln k + \frac{1}{2}\ln L} = e^c \implies e^{\frac{1}{2}\ln k} \cdot e^{\frac{1}{2}\ln L} = e^c$

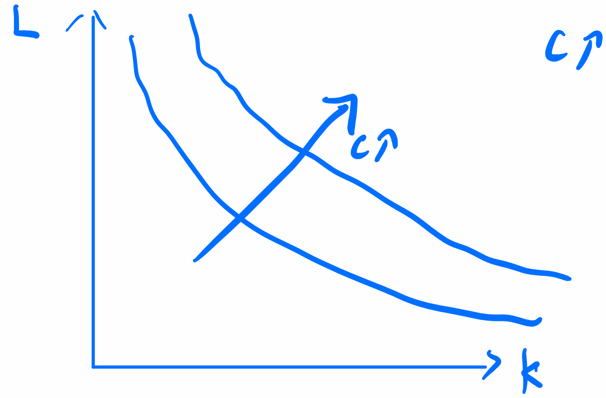
exponentiate with base e

$(e^{\ln k})^{\frac{1}{2}} \cdot (e^{\ln L})^{\frac{1}{2}} = e^c$

$(k)^{\frac{1}{2}} \cdot (L)^{\frac{1}{2}} = e^c$

$k \cdot L = e^{2c} \implies L = \frac{e^{2c}}{k}$


4
3
2
1
1
2
3
4
5
K
c ↑


L
k
c ↑
c ↑

Tutorial Q4: Isoquants

Knowledge used

- Isoquants are level curves of a production function.
- A level curve of level C solves $Q(K, L) = C$.

Solution

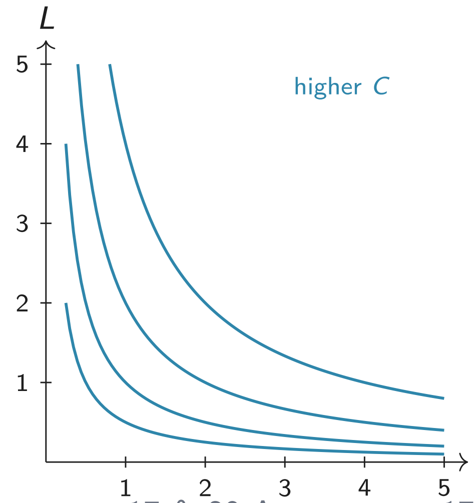
(a) For $Q(K, L) = K^{1/2}L^{1/2}$, the level curve of level $C > 0$ satisfies

$$K^{1/2}L^{1/2} = C \iff KL = C^2 \iff L = \frac{C^2}{K}.$$

(b) For $Q(K, L) = 0.5 \ln K + 0.5 \ln L$, the domain is $K, L > 0$. The level curve of level C satisfies

$$0.5 \ln K + 0.5 \ln L = C \iff K^{1/2}L^{1/2} = e^C \iff L = \frac{e^{2C}}{K}.$$

Hence the contour plot has the same shape as part (a).



Tutorial Q5

6 min

Question 5

(a) Write down the formula for a linear function g with domain $\mathbb{R}^2$ and codomain $\mathbb{R}$ that has the property

$$g(7,2) - g(3,2) = 8, \qquad g(7,5) - g(3,2) = 5.$$

(b) Consider $\widehat{g} : \mathbb{R}^2 \to \mathbb{R}$, $\boldsymbol{x} \mapsto g(\boldsymbol{x}) + 2$, where g is the linear function from part (a). How does the value of $\widehat{g}$ change if $\boldsymbol{x}$ changes

(i) from $(-3, 0)$ to $(2, 1)$?

(ii) from $(2, 5)$ to $(1, 3)$?

$a = (a_1, a_2)$

$x = (x_1, x_2)$

(a) $g : \mathbb{R}^2 \to \mathbb{R}$, $g(x) = a \cdot x = a_1 \cdot x_1 + a_2 \cdot x_2 = 2x_1 - x_2$

$g(7,2) = 7a_1 + 2a_2$

$g(3,2) = 3a_1 + 2a_2$

$g(7,5) = 7a_1 + 5a_2$

$$\Rightarrow \begin{cases} g(7,2) - g(3,2) = 8 \\ (7a_1 + 2a_2) - (3a_1 + 2a_2) = \underline{4a_1 = 8} \Rightarrow a_1 = 2 \\ g(7,5) - g(3,2) = 5 \\ (7a_1 + 5a_2) - (3a_1 + 2a_2) = \underline{4a_1 + 3a_2 = 5} \Rightarrow 8 + 3a_2 = 5 \end{cases}$$

$a_2 = -1$

(b) $\hat{g}: \mathbb{R}^2 \to \mathbb{R}$, $\hat{g} = g + 2 = 2x_1 - x_2 + 2$

Change formula: $\Delta \hat{g} = a \cdot \Delta x = a_1 \cdot \Delta x_1 + a_2 \cdot \Delta x_2$ $\quad a = (2, -1)$

$= 2 \cdot \Delta x_1 - 1 \cdot \Delta x_2$

(i) from $(-3, 0)$ to $(2, 1)$?
(ii) from $(2, 5)$ to $(1, 3)$?

(i) $(-3, 0) \to (2, 1)$

$\Delta x_1 = 2 - (-3) = 5$

$\Delta x_2 = 1 - 0 = 1$

$\Delta \hat{g} = 2 \cdot 5 - 1 \cdot 1 = 10 - 1 = 9$

(ii) $(2, 5) \to (1, 3)$

$\Delta x_1 = 1 - 2 = -1$

$\Delta x_2 = 3 - 5 = -2$

$\Delta \hat{g} = 2 \cdot (-1) - 1 \cdot (-2) = -2 + 2 = 0$

Tutorial Q5(a): Finding the Linear Function

Knowledge used

Use the lecture definition of a linear function:

$$g(\boldsymbol{x}) = \boldsymbol{a} \cdot \boldsymbol{x} = a_1 x_1 + a_2 x_2.$$

For linear functions, differences depend only on the change in the input.

Solution

Let $g(\boldsymbol{x}) = a_1 x_1 + a_2 x_2$. The two conditions give

$$g(7,2) - g(3,2) = 4a_1 = 8 \quad \Rightarrow \quad a_1 = 2,$$

$$g(7,5) - g(3,2) = 4a_1 + 3a_2 = 5 \quad \Rightarrow \quad 8 + 3a_2 = 5 \quad \Rightarrow \quad a_2 = -1.$$

Therefore,

$$g(\boldsymbol{x}) = (2, -1) \cdot \boldsymbol{x} = 2x_1 - x_2.$$

Tutorial Q5(b): Changes in the Affine Function

Knowledge used

For an affine function $\widehat{g}(\boldsymbol{x}) = \boldsymbol{a} \cdot \boldsymbol{x} + b$, a change $\Delta\boldsymbol{x}$ gives

$$\Delta y = \boldsymbol{a} \cdot \Delta\boldsymbol{x}.$$

The constant term does not affect changes.

Solution

Here $\boldsymbol{a} = (2, -1)$, so $\Delta y = (2, -1) \cdot \Delta\boldsymbol{x}$.

(i) From $(-3, 0)$ to $(2, 1)$:

$$\Delta\boldsymbol{x} = (5, 1),$$

$$\Delta y = (2, -1) \cdot (5, 1) = 9.$$

(ii) From $(2, 5)$ to $(1, 3)$:

$$\Delta\boldsymbol{x} = (-1, -2),$$

$$\Delta y = (2, -1) \cdot (-1, -2) = 0.$$

Key Takeaways

- Level curves solve $f(\boldsymbol{x}) = c$ and contour plots collect several level curves.
- Continuity can often be proved using sums, products, quotients, compositions, projections, and componentwise arguments.
- Linear and affine functions convert changes in variables into changes in values through the dot product.